Technology Stack

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Comprehensive overview of programming languages, machine learning frameworks, libraries, and runtime architecture powering CrimeVision AI.

Layer / Domain	Technology / Framework	Version / Package	Role in CrimeVision AI Platform
Core Language	Python	3.10 / 3.11 / 3.13	Backend API server, ML inference orchestration, data processing.
Web Framework	Flask & Flask-CORS	3.0+	Modular Blueprint routing, REST API endpoints, static UI serving.
Deep Learning	PyTorch & Transformers	2.2+ / 4.38+	Model execution engine for RT-DETR, CLIP, BLIP-2, and Whisper.
Computer Vision	OpenCV (cv2) & Pillow	4.9+ / 10.2+	Video frame extraction at 1Hz, image preprocessing, bounding box rendering.
Speech & Audio	OpenAI Whisper	whisper-base	Multilingual speech-to-text transcription with timestamp segmenting.
LLM Reasoning	Groq Cloud SDK	LLaMA-3.3-70B	Forensic scenario deduction, cross-lingual translation, chatbot queries.
Frontend UI	HTML5, CSS3, ES6 JS	Vanilla ES6+	Cyberpunk forensic HUD, reactive tabs, asynchronous API fetch clients.
3D Visualization	Three.js & WebGL	r128 / OrbitControls	Interactive 3D crime scene rendering, camera rotation, 3D evidence cubes.
PDF Generation	ReportLab Platypus	4.4.10	Court-admissible PDF synthesis with table formatting and XML sanitation.